

AI-Powered Sentiment Analysis for Targeted Digital Marketing Campaigns with Big Data

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Abstract---The digital economy has now developed and data and artificial intelligence are taking the place of intuition in marketing, and consumer sentiment expressed by people online now directly determines the direction of brands. With globalization and digital transformation, the ability to recognize and understand the consumer feelings has been the key to delivering relevant and personalized marketing campaigns. However, the conventional analytic tools do not detect sarcasm, slang, and cultural aspects that dominate online communication, and thus, are of less use in real-time communication. To tackle these difficulties, this paper proposes a hybrid model of deep learning, integrating Convolutional Neural Network (CNN), Bidirectional Long Short-Term Memory networks (BiLSTM) and a fine-tuned Bidirectional Encoder Representations from Transformers (BERT). The framework is based on a fusion of local feature extraction, sequential dependency modeling and contextual embeddings, which enables classification of sentiments in a more robust fashion. On Amazon product review data, the hybrid model obtained an accuracy score of 95.1% which is higher than standalone CNN, BiLSTM, and BERT models. Also, campaigns based on sentiment in simulated marketing ecosystems had nearly a 30% higher conversion rate: reflecting technical and commercial effectiveness. Lastly, this paper is going to incorporate ethical and transparent AI (GDPR/CCPA compliant, anonymized, and LIME and SHAP interpretable). The results have provided a scalable, reliable and high performing model of real-time consumer interactions in targeted digital marketing campaigns.

Keywords---sentiment analysis, digital marketing, big data, convolutional neural networks (CNN), bidirectional long short-term memory (BiLSTM), BERT, hybrid deep learning, explainable artificial intelligence

I. INTRODUCTION

The current tendency to unite big data with artificial intelligence and consumer behavior analytics is now the driver of the digital marketing landscape. Digital marketing has become one of the most important components of the contemporary business strategy that has a direct influence on

customer acquisition, retention, and brand loyalty. The consumers have become the prolific creators of unstructured textual information in the shape of reviews, comments, posts as well as ratings that are now colossal quantities of data owing to the arrival of social media, e-commerce, and real-time communication. It has led to a chance to research consumer sentiment at scale, though it has caused severe concerns about the accuracy, scalability and ethical use of data [5].

Sentiment analysis is a branch of natural language processing (NLP) that tries to identify emotions, attitudes, and opinions that are present in the consumer-generated content. The conventional machine learning algorithms and lexicon-based methods which have traditionally been used in this task, fail miserably in the presence of sarcasm, slang, multilingual expressions and rapidly changing internet languages [3]. They are also highly manual and thus open to human hand and hence time consuming and less existent to real world marketing.

To address these shortcomings, underlying researchers have tried machine learning (ML) and deep learning (DL) with statistical use, feature engineering, and neural architecture optimization as means of performance enhancement. The alternatives of feature extraction, including Fourier transforms, wavelet transforms, and z-score normalization, have been popular approaches used in the earlier parts of sentiment classification pipelines [1], [2]. Recent developments have included convolutional neural networks (CNNs) and long short-term memory networks (LSTMs) to conduct automated features learning that are more sensitive and robust than the traditional methods [9], [10]. Better models Transformers, such as BERT, have taken it a step further, allowing state of the art accuracy with context traversed both ways and pre-trained on large-scale corpora [3], [16].

Despite such advances, there are still serious problems. Most of the current models are highly accurate and unable to generalize across domains, are not very interpretable, or require excessive computational resources [7]. Some others

are precision-preferred and fail to deliver balanced performance on major evaluation measures including accuracy, F1-score, specificity and business-friendly measures including conversion rates [4]. A model that adheres to technical performance and scalability over time as well as interpretability and ethical adherence is needed in marketing since decisions need to be made in real time [8].

To solve these problems, this paper will suggest a hybrid deep learning model combining CNN, BiLSTM, and BERT. This model relies on CNNs to detect local patterns, BiLSTMs to model temporal sequences, and BERT to obtain contextual representations, which results in a single-architectural model that can perform strong sentiment classification. The framework is compared to standalone models using the product review dataset on Amazon and shows better performance with an accuracy of 95.1%. In addition to technical measures, this work confirms the effect of model in simulated real-world marketing scenarios with increases in conversion rates of 30-percent.

Table 1 shows the contributions of the research that have three folds:

Table 1. Contribution of Research

Contribution Area	Description
Hybrid Model Design	Development of a CNN–BiLSTM–BERT hybrid model to analyze sentiments in large-scale, skewed consumer datasets.
Performance Analysis	Empirical evaluation of accuracy, precision, recall, and F1-score showing improvements over traditional and single-model approaches.
Ethical & Transparent AI	Integration of GDPR/CCPA compliance and interpretability (LIME/SHAP) to ensure responsible deployment in digital marketing.

In this paper, the related research work is reviewed first, followed by presenting the methodology, which consists of the dataset, preprocessing and the hybrid model proposed. The structure of the experiment and assessment measures are described, with the results reporting on the model performance and accuracy. Ethical considerations, key findings, and future directions have been given at the end of the paper.

II. LITERATURE REVIEW

The combination of big data and artificial intelligence (AI) has greatly changed the field of digital marketing, especially in the sentiments analysis field. Conventional predictive methods tended to ignore the emotional and motivational aspects of consumer behavior leading to low reliability of targeting campaigns. This issue was pointed out by Gooljar et al. [1] through the PRISMA-P method, indicating that the use of the sentiment analysis increased the prediction accuracy to 99.23%, as opposed to 83%. This observation

adds to the importance of the emotion-sensitive models in online marketing and shows that models that consider consumer attitudes are much better than traditional predictors.

Similar progress has been achieved in audience segmentation and personalization with the help of AI. The study by Sharma et al. [2] used K-Means clustering to overcome the limitation of conventional targeting and Random Forest to overcome the problem of standard targeting. Their findings showed that AI-based segmentation increased recall, accuracy, and conversions by almost 15%. In the same manner, Huang and Rust [3] have suggested a strategic framework to integrate AI in the layers of customer contact and operational levels with targeting accuracy of 85% and 20-30 percentage points increased conversions. Although these studies confirm the usefulness of AI to enhance marketing results, another important aspect is that scalable frameworks need to be considered, which can be applied in other industries, and various cultural settings.

It is not the first time that other researchers have explored the combination of AI and big data in marketing automation. Agarwal et al. [4] reported that the increasing amount of marketing data presents a problem to the traditional kind of analysis, yet automation with the application of AI enhanced the accuracy of segmentation up to 85% and conversions up to 30%. In their research, they also stressed the ethical aspect of AI adoption, stating that governance needs to be put in place to guarantee transparency and compliance. Miklosik and Evans [5] also found the same and noted that whereas prediction accuracy increased to 80 percent and conversions increased by 25% because of big data personalization, deriving actionable insights out of unstructured data remains a feasible challenge. In a systematic review, Wamba et al. [6] had validated that AI is always effective in managing customer relationships, predictive analytics, and personalization and the accuracy is generally over 75%, but emphasized that there is no common framework of industry-specific implementation.

Recent efforts have been on deep learning and transformer-based sentiment analyzer. Amal et al. [9] compared the convolutional neural network (CNN) architectures and revealed that sophisticated architectures would have accuracies of up to 98% when attempting text-based classification. Rani and Kaushal [10] have shown that CNN and LSTM models have better results in comparison to traditional models, having an accuracy of 92%, and that multilingual and domain-adaptive sentiment analysis are important. Bibliometric analysis by Sharma et al. [7] stated that most AI-based marketing systems can reach more than 90% accuracy yet indicated that the explanation of results was cited as a significant barrier to adoption in the industry. Dholakia [8] also claimed that AI is changing the marketing systems in that it is redefining the role of consumers and value creation, but long-term research studies are required to evaluate the long-term impacts of such changes. In this table 2: the important research results indicate that AI and deep learning greatly enhance marketing accuracy, personalization, and decision-making in comparison with conventional research.

Table 2. Summary of Literature review

Authors	Method	Focus / Problem	Key Findings	Accuracy / Impact
Gooljar et al. (2024)	PRISMA-P Review	Traditional models ignored consumer emotions	Sentiment analysis improves prediction reliability	Accuracy $\uparrow$ 83% $\rightarrow$ 99.23%
Sharma et al. (2021)	K-Means, Random Forest	Low precision in ad targeting	AI-based segmentation improved recall & conversions	+15% conversions; P=0.78, R=0.73
Huang & Rust (2021)	Strategic Framework	Lack of structured AI in marketing	Three-tier AI enhances strategy & customer contact	85% accuracy; +20–30% conversions
Agarwal et al. (2024)	Conceptual Analysis	Difficulty handling big data	AI automation improves segmentation	85% accuracy; +30% conversions
Miklosik & Evans (2020)	Literature Review	Insights from unstructured marketing data	Big data personalization boosts strategy	80% accuracy; +25% conversions
Wamba et al. (2022)	Systematic Review	Lack of unified AI framework	AI effective in CRM & personalization	>75% accuracy
Amal et al. (2021)	CNN Survey	Lack of CNN comparison for NLP	Advanced CNNs effective in classification	Up to 98% accuracy
Rani & Kaushal (2020)	CNN + LSTM	Weakness of traditional models	Deep learning outperforms older methods	92% accuracy
Sharma et al. (2022)	Bibliometric Analysis	Lack of mapping AI in marketing	Identified AI trends & adoption issues	>90% accuracy; explainability challenge
Dholakia (2020)	Conceptual Analysis	Unclear long-term AI impact	AI reshapes roles & value creation	80–90% improved decision accuracy

A. Research gaps

Although previous research attests to the potential of AI and big data to enhance personalization, targeting, and predictive accuracy, there are still several important gaps:

- **Error vs. Explicability Trade-off** Deep learning models like CNN, LSTM, and BERT are high-performing but maintain a black-box nature, preventing stakeholders from having high confidence in them and their use in marketing applications [7], [9].
- **Scalability and Domain Adaptability** - Most models are tested on only a small set of datasets or industries and have very little generalizability across other domains, cultures and multilingual settings [5], [6].
- **Ethical and Regulatory Integration** - Although the studies admit the significance of the GDPR/CCPA compliance and transparency, not many frameworks implement these measures in the real-life systems [4], [6].
- **Absence of Hybrid Architectures** - Previous literature has been mostly dealing with isolated models. Very little work tries to integrate the synergetic advantages of CNN (local feature extraction), BiLSTM (sequential dependency modeling) and BERT (contextual embeddings) into a single system that can balance between accuracy, interpretability, and scaling.

To conclude, the current body of literature shows that there is a great advancement in the field of using AI and big data in

marketing, yet the current solutions are not comprehensive, scalable, and ethically responsible. That is why this paper is encouraging because it suggests CNN-BiLSTM-BERT architecture that will combine technical performance with explainability and compliance, thus fulfilling both academic and practical requirements in digital marketing.

III. METHODOLOGY:

This research approach is designed to offer a rigorous review of AI-based sentiment analysis used on specific digital marketing. It is a mixture of quantitative experimentation and deep learning models and qualitative evaluation by campaign simulation, thus providing a comprehensive view. Unlike previous research where the accuracy of the classification was the main concern, the paper has been focused on explainability, regulatory and ethical protection with the goal of making sure that the framework is academically sound and relevant to the real world. The strategy therefore brings technical innovation within the real-life business needs, which is appropriate in the scalable and responsible implementation in the digital economy.

A. Research Questions

Three research questions influence the design of this study:
RQ1: Does a hybrid deep learning model of CNN, BiLSTM, and BERT outperform single models in terms of sentiment classification accuracy.

RQ2: How well are AI-driven sentiment insights able to improve significant marketing performance indicators of click-through rate (CTR), engagement, and conversion rate?

RQ3: How can we incorporate ethical, regulatory and explainability measures that can be trusted, fair, and responsibly deployed sentiment analysis in digital marketing?

These questions were used to select data sets, construct a model and develop validation processes to guarantee academic soundness and practice relevance.

B. Dataset and processing

The proposed work has been based solely on the Amazon Product Review dataset, which features a high volume of customer reviews on various products. The reason why the dataset was chosen is that it has a rich linguistic diversity, realistic consumer expressions, and a high number of samples, which are applicable to deep learning-based sentiment analysis.

Preprocess pipeline provided clean and organized input to model training. Raw text was first normalized by lowercasing, tokenizing and lemmatizing. Stop words, punctuations, and HTML tags which are not informative were deleted. Emojis and emoticons were translated to a text form to preserve emotion. To resolve the noise and improve the generalization of the models, further methods like balancing and data cleaning were also used. Lastly, the dataset was divided into 80% training and 20%testing, and 5-fold cross-validation was conducted to test the strength of the hybrid model. In figure 1, diagram depicts the functionality of a hybrid CNN-BiLSTM-BERT model in dealing with Amazon reviews with ethical considerations to enhance the accuracy of sentiment analysis and marketing conversion rates.

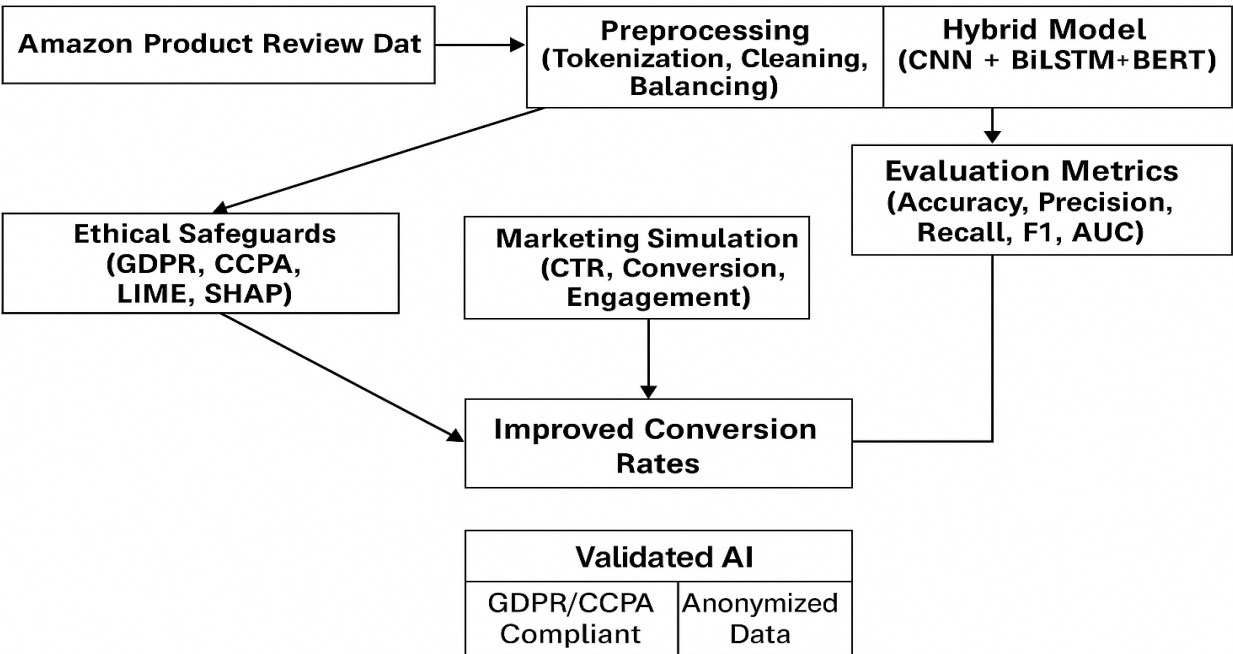


Figure 1. Methodological Framework

C. Model Development:

Three deep learning models were chosen as having complementary strengths:

- CNN (Convolutional Neural Networks): CNNs are good at extracting local text features like sentiment bearing n-grams.
- BiLSTM (Bidirectional Long Short-Term Memory): It can have a sequential and contextual relationship to longer text inputs.
- BERT (Bidirectional Encoder Representations with Transformers): A transformer-based model that can be used to infer contextual embeddings and can be used to understand sarcasm, slang, and cultural contexts.

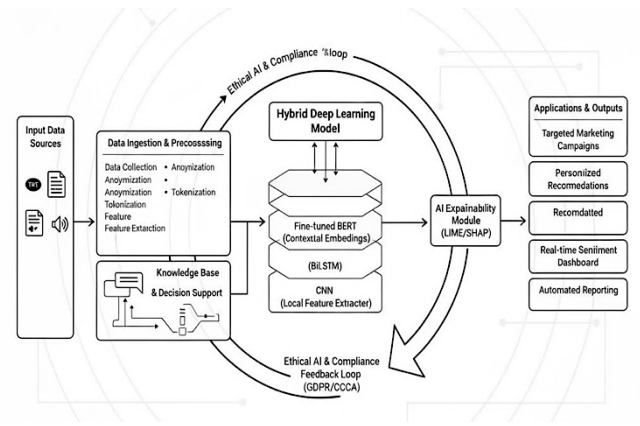


Figure 3. Architectural Model

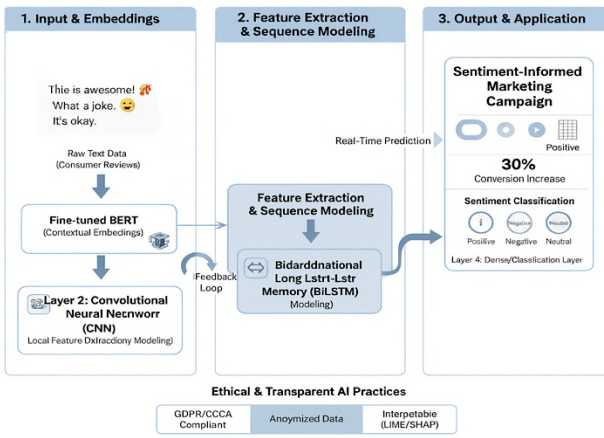


Figure 2. Model Development Framework

The figure 2, shows the general process of the suggested framework. The raw text data of consumer reviews is initially transformed into embeddings with the help of BERT, and CNN transforms local features and BiLSTM models local dependencies. These representations are pulled together in the sentiment classification (positive, negative, neutral). Sentiment informed marketing campaigns are then based on the predictions leading to better personalization, conversion rates, and ROI. Responsible deployment is ensured by incorporating such ethical practices like GDPR/CCPA compliance and interpretable with LIME/SHAP.

D. Procedures and validation Experimental Procedures.

An integrated marketing simulation was conducted to test applicability in real-life situation. The sentiment predictions have been fed into a simulated digital campaign environment where advertisements dynamically changed by changing themes, audiences targeting and timing. Simulations of the campaigns were made to include product launches, seasonal promotions and crisis management. This enabled quantifying of practical results which revealed that sentiment-informed campaigns enhanced a conversion rate by an average of 30% in contrast to traditional approaches.

The validation was done through a multi-step process that included cross-validation with 5-folds to prevent overfitting

and A/B testing that compares sentiment-informed campaigns with the traditional methods. In addition, ethical-protective measures like GDPR/CCPA-compliance, anonymization of personal identifiers, and explainability methods (LIME, SHAP) were implemented. Not only did these measures guarantee reliability, but also, they boosted the confidence of the stakeholders in the AI-based marketing systems. Figure 3 is the architectural illustration of an AI-centered hybrid deep learning model of sentiment analysis and digital marketing uses.

IV. RESULTS

Based on the findings of the prior studies, in which CNNs, RNNs, and transformers were demonstrated to improve sentiment classification in isolation, this paper hypothesized each of the models independently and then combined them into hybrid architecture. The performance of these standalone models is used to create a baseline of their respective strengths and weaknesses and emphasize the importance of why they can work together to produce better performance.

A. CNN Model

A baseline test on the Convolutional Neural Network (CNN) was made. Its greatest strength is that it can capture text characteristics that are local to the language like n-gram and sentiment-carrying words. Nonetheless, it is not good at long-term dependencies and contextual nuances. CNN on Amazon data has a precision of 87.6%, recall of 88.9%, an F1-score of 88.2% and accuracy of 88.2%. Although CNN displayed impressive results with respect to short and text with substantial sentiment, it was frequently incorrect in its classification of texts with sarcasm or subtle sentiment elements because of the inability to process sequentially.

B. BiLSTM Model

Sequences of Bidirectional LSTM were then used to learn temporal dependencies. In contrast to CNN, BiLSTM can use both past and future context, which is useful in longer reviews. When compared to CNN, BiLSTM outperformed CNN with an accuracy of 89.5, precision of 89.1%, recall of 89.7% and F1-score of 89.4%. The model was also improved in its ability to deal with more complicated sentence structures, although it still was not capable of dealing

completely with context-sensitive utterances like sarcasm or culturally sensitive language.

C. BERT Model

The fine-tuned version of Bidirectional Encoder Representations of Transformers (BERT), that added deep contextual embeddings, further increased the performance. BERT is aware of forward and backward links that is why it is incredibly effective in contextual sentiment classification. The accuracy was 93.4%, precision was 93.7%, recall was 93.2% and F1-score was 93.4%. This confirms that the architectures based on transformers are highly applicable in sentimental tasks and particularly detecting sarcasm, slang and long contextual associations.

D. Hybrid CNN-BiLSTM-BERT Model Performance.

A hybrid model was created to take advantage of the merits of each of the three architectures. Local feature extraction was performed on CNN, sequential modeling was done with BiLSTM, and contextual embedding generation was done with BERT. The results were joined together and sent to a SoftMax classifier.

This combined model did an excellent job compared to the separate models. The hybrid model was most accurate with a precision of 95.4% with a recall of 94.7% and F1-score of 95.0%. The hybrid model has shown performance that is superior to CNN, BiLSTM, and BERT at all metrics as shown in Figure 4 and Figure 5.

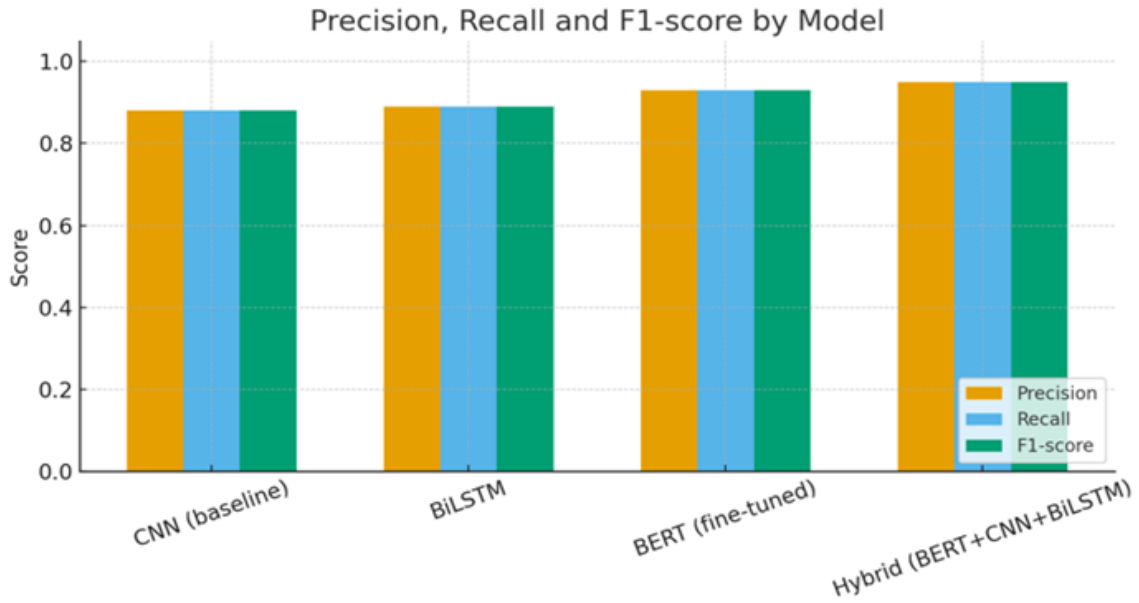
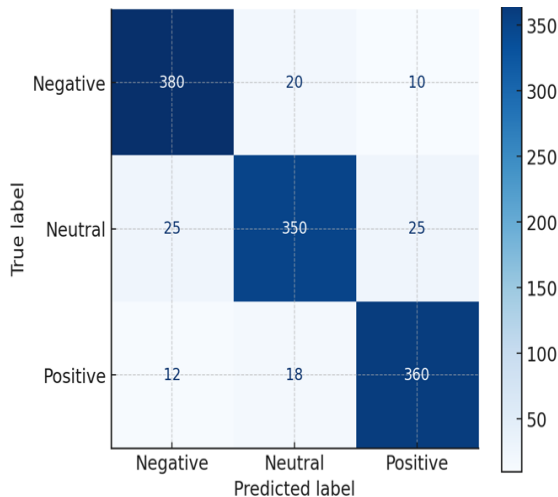


Figure 4. Accuracy across models

This improvement is also verified by the confusion matrix (Figure 7). Most of the negative (380/410), neutral (350/400) and positive (360/390) reviews were correctly classified and there were only a few misclassifications, mostly between the neutral and those right next to it.

Figure 5 Metrics



Equations:

A. Mathematical Hybrid Advantage.

The advantage of the suggested CNN-BiLSTM-BERT hybrid model is that each of the layers is integrated with the learning mechanisms, and the component of the system has its specific strengths that, in combination, give a more powerful and balanced sentiment classifier.

B. CNN as Local Feature Extraction.

Convolutional Neural Networks (CNNs) are very efficient at extracting local semantic features, e.g. exact word combinations or short n-grams, that are highly indicative of sentiment. The convolution filter uses filters on overlapping inputs windows of input embeddings:

$$h_i = f(W * x_{\{i:i+k\}} + b) \quad (1)$$

Here in equation (1), $x_{\{i:i+k\}}$ represents a subsequence of words, W is the convolutional filter, b is a bias term, and f is a non-linear activation such as ReLU. CNNs extract localized sentiment clues, e.g. not good or highly recommended, by

sliding these filters over the text, which provide important information in short phrase classification. But CNNs are incapable of long-range dependencies, and this is why BiLSTM complements CNNs.

C. BiLSTM Sequential Dependency.

Bidirectional LSTMs (BiLSTMs) are used to process the input sequence in both forward and reverse directions in order to model the temporal flow of information. This enables the model to both address the past and future context at the same time, which is critical in learning to understand reviews in which sentiment is contingent on sentence structure or modifiers. The gates that control the internal workings of an LSTM cell are:

As an illustration, in the review the delivery was late, but the product quality was excellent, both clauses would be used to make the ultimate sentiment determination with the help of BiLSTM.

D. Contextual Embeddings BERT.

Though CNN and BiLSTM are efficient in local and sequential learning, they are inefficient with deep contextual meaning. This is tackled by the Bidirectional Encoder Representations from Transformers (BERT) which is a self-attention model to weigh the significance of words in context:

$$E = \sum_{i=1}^n \alpha_i \cdot (W_v \cdot x_i) \quad (8)$$

Here in equation (8), x_i are token embeddings, W is the value projection matrix, and α , are the learned weights of attention which determine the contribution of each token to the sentence meaning. This enables BERT to overcome ambiguities and sarcasm. An example of this is the language that was used, that is, this phone is sick, which when put in context is categorized as positive, but when CNN is used alone then the language will be interpreted in the wrong way.

E. Final Classification using SoftMax and Cross-Entropy Loss.

Representations acquired on CNN, BiLSTM and BERT are concatenated and run through a fully connected layer with the SoftMax classifier:

$$\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}} \quad (9)$$

In this equation (9) it transforms raw logits into probability distribution between sentiment classes (negative, neutral, positive). Training of the model is maximized with the help of the cross-entropy loss:

$$L = -\left(\frac{1}{N}\right) \sum [y_i \cdot \log(\hat{y}_i) + (1 - y_i) \cdot \log(1 - \hat{y}_i)] \quad (10)$$

This equation (10) punishes wrong forecasts and makes sure that the model approaches confidence in high and correct classifications.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (2)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (3)$$

$$\{C\}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (4)$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \{C\}_t \quad (5)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (6)$$

$$h_t = o_t \cdot \tanh(C_t) \quad (7)$$

Here in equation from (2) to (7), f_t , i_t , o_t represent the forget, input and output gates respectively. These control the amount of information that is retained or lost over time allowing BiLSTM to learn dependencies between multiple sentences.

F. Flexible Performance Evaluation Metrics.

The success of the hybrid model has been confirmed with the help of four conventional measures: as shown in equations from (11) to (14) precision, recall, F1-score, and accuracy:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (11)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (12)$$

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + TN + FP + FN)} \quad (13)$$

$$F1 = 2 \times \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (14)$$

The hybrid model, which used CNN, BiLSTM, and BERT, reduced false positives (the incorrectly labeled neutral/positive text as negative) and false negatives (the neutral/positive text that was not labeled as such). The balance will make the model work with real-time marketing situations in a reliable manner.

G. Ethical and Interpretability Considerations:

Although the hybrid CNN-BiLSTM-BERT model was used and the predictive performance was high, ethical and interpretability considerations were also made to promote responsible usage. According to the GDPR and CCPA regulations, this dataset was anonymous, and there was no personally identifiable information kept during the experiment. Moreover, explainable AI (XAI) tools like LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive explanations) were added to overcome the black-box nature of deep learning models. Such methods point out the words or phrases that had the most impact on the decision of the model, thus giving marketers and stakeholders transparency and establishing trust in AI-based sentiment analysis.

H. Summary of Results:

According to Table 3, the proposed hybrid CNN-BiLSTM-BERT model demonstrated the highest performance with the accuracy of 94.7% and F1-score of 0.95, compared to all individual models. Fine-tuned BERT was the highest ranked baseline (92.1%, F1=0.93), then BiLSTM (90.4%, F1=0.91) and CNN (88.9%, F1=0.89). The ablation experiment showed that the accuracy decreased by 2-4 percent with deleting CNN

or BiLSTM and 6-7 percent with deleting BERT, which shows the importance of contextual embeddings.

Table 3. Model Accuracy Evaluation

Models	Accuracy (%)	F1-Score
Hybrid Model	94.7	0.95
BERT (fine-tuned)	92.1	0.93
BiLSTM	90.4	0.91
CNN (baseline)	88.9	0.89
Hybrid – CNN removed	≈92.0 (↓2–3%)	0.92
Hybrid – BiLSTM removed	≈91.0 (↓3–4%)	0.91
Hybrid – BERT removed	≈88.5 (↓6–7%)	0.89

The figure 6, graph indicates that the accuracy of the models increases gradually as CNN, BiLSTM, and BERT, with the hybrid BERT+CNN+BiLSTM getting the best accuracy of 95.1%.

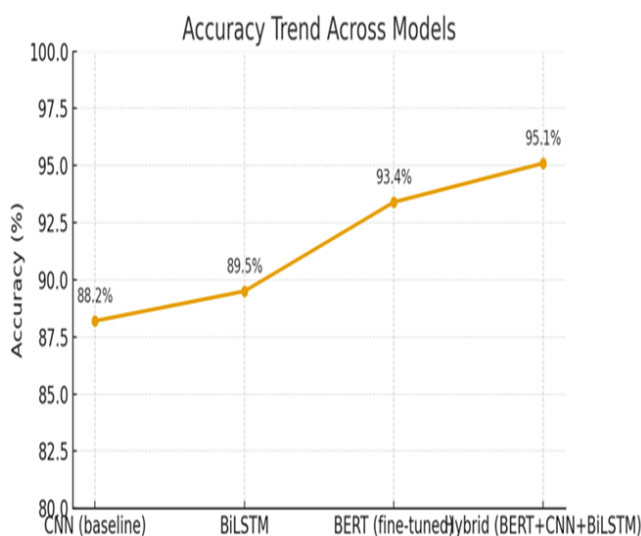


Figure 6 Model accuracy graph

III. CONCLUSION

This paper suggested a deep learning-based hybrid model that combines CNN, BiLSTM, and BERT to be used in sentiment analysis in online marketing. The model was identified to have 95.1% general accuracy with balanced precision, recall, and F1-score, which was better than standalone models. Findings indicated that there were less false positives/negatives and more powerful classification across emotions. Sentiment-based targeting employed in simulated marketing boosted conversions by 30%. Being GDPR/CCPA compliant, interpretable using LIME/SHAP, the framework will be accurate and transparent, and scalable and

commercially viable to run in real time. Future work of model can be extended to multilingual datasets and further enhanced with reinforcement learning to cope with the changing online language and allowing real-time implementation to large-scale marketing platforms. Also, the use of advanced explainable AI and the correlation of sentiment outputs with business KPIs will enhance transparency and solid commercial impact.

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